Multimodal Biosensor Patch with Sphere-Shaped NPC-Supported $Ti_3C_2T_x$ Heterostructures on Laser-Induced Graphene for Wearable Sweat Monitoring

Ahmad Abdus Samad, Jiwon Go, Jae Yeong Park*
Kwangwoon University, Department of Electronic Engineering
E-mail: jaepark@kw.ac.kr

웨어러블 땀 모니터링을 위한 레이저 유도 그래핀 지지 구형 나노다공성 카본 지지 Ti_3C_2Tx 맥신 이종 구조 다중모드 바이오센서 패치

아부다스 사마드, 고지원, 박재영*,
광운대학교, 전자공학과
E-mail: jaepark@kw.ac.kr

Abstract

This study presents a flexible electrochemical sensor patch for sweat analysis, designed to monitor key biomarkers including glucose and uric acid. The sensor integrates 3D-NPC@2D-$Ti_3C_2T_x$ onto three-dimensional laser-induced graphene (LIG) electrodes fabricated via a CO_2 laser. The resulting 3D/2D heterostructured transducing layer facilitates efficient electron transfer, suppresses MXene restacking, and enhances interfacial stability while providing abundant exposed active sites. Owing to these synergistic effects, the sensor achieves high sensitivities of 168 and 226 $\mu AmM^{-1}cm^{-2}$ for glucose and uric acid detection, respectively. To the best of our knowledge, this work represents the first demonstration of 3D-NPC@2D-$Ti_3C_2T_x$ heterostructures employed as a transducing layer on LIG for sweat monitoring, offering a promising strategy for comprehensive sweat biomarker analysis.

Keywords: ***2D nanomaterials (2D 나노소재), Sphere-shaped nanoporous carbon (구형 나노다공성 탄소), LIG(레이저 유도 그래핀), Wearable(웨어러블)***

1. Introduction

Wearable sweat sensors have emerged as powerful tools in personalized healthcare due to their ability to non-invasively access molecular-level physiological information through continuous sweat analysis[1]. Simultaneous monitoring of key sweat biomarkers like glucose, uric acid, and sodium ions (Na^+) provides valuable insights into renal function and metabolic dysregulation in diabetic patients[2]. However, the practical implementation of sweat-based biosensing remains challenging because of the intrinsically low concentrations of glucose and uric acid in sweat. Enhancing detection sensitivity therefore critically depends on advanced transduction materials, such as MOF-derived nanoporous carbons (NPCs) and two-dimensional nanomaterials including MXenes[3]. Nevertheless, MOF pyrolysis often reduces hydrophilic functional sites and induces NPC aggregation during dispersion, while MXenes suffer from chemical instability and nanosheet restacking, which limit their electrochemical performance. Herein, we report a three-dimensional sphere-shaped NPC-supported $Ti_3C_2T_x$ heterostructure as an efficient transducing layer for high-performance sweat biosensing. This rationally engineered 3D@2D architecture effectively suppresses MXene restacking and NPC agglomeration, while simultaneously increasing accessible active sites and improving chemical stability. The 3D sphere-shaped NPC framework is derived from a transition-metal-based bimetallic Co@Ni-MOF.

2. Preparation of materials

2.1 Synthesis of NPC:

The Co@Ni-MOFs were synthesized via a modified solvothermal method. Briefly, $Co(NO_3)_2{\cdot}6H_2O$ and $Ni(NO_3)_2{\cdot}6H_2O$ were dissolved in N, N-dimethylformamide (DMF) to form a homogeneous metal precursor solution. Separately, 1,4-benzenedicarboxylic acid (BDC) was dissolved in DMF. The two solutions were then combined and magnetically stirred for 30 min at room temperature. The resulting mixture was transferred into a Teflon-lined stainless-steel autoclave and subjected to solvothermal treatment at 180°C for 24 h. After that, the precipitated product was collected by centrifugation, thoroughly washed with DMF and DI to remove unreacted species, and dried in a vacuum oven at 60°C for 12 h to obtain the Co@Ni-MOF precursor. Then the dried Co@Ni-MOF was calcined at 900°C for 5 h under a nitrogen atmosphere with a heating rate of 5°C min^{-1}, yielding a three-dimensional Co@Ni@Carbon material. The black powder was then mixed with a 10% hydrofluoric acid solution while stirring for 24 hours. After being etched, the product was thoroughly washed with deionized water until the pH level was neutral. Finally, the collected sample was dried in a vacuum oven at 60°C overnight. The resulting material, consisting of nanoporous carbon with residual nickel species, was denoted as Ni-doped NPC.

2.2 Synthesis of MXene:

Multilayer Ti_3CT_x MXene was synthesized by selectively etching Ti_3AlC_2 (MAX phase) using a mixed acid solution of HCl, HF, and deionized (DI) water. Then few-layer Ti_3CT_x MXene was obtained by intercalating the multilayer MXene with LiCl, followed by stirring for 24 h at 35 °C. The dispersion was centrifuged, and the dark supernatant was collected and further centrifuged for 1 h. The resulting MXene was redispersed in DI water and washed through repeated centrifugation cycles. Finally, the few-layer $Ti_3C_2T_x$ MXene was dried overnight in a vacuum oven at 60 °C.

2.3 Preparation of heterostructured 3D-NPC@2D-$Ti_3C_2T_x$:

Sphere-shaped 3D NPC and 2D MXene were hybridized via a combined sonication-stirring process. Briefly, NPC and MXene were mixed at a mass ratio of 2:1 and dispersed in ethylene glycol (EG), through probe sonication for 60 min to achieve uniform dispersion and intimate interfacial contact. The resulting dispersion was then vigorously stirred at 60 °C for 24 h to facilitate interfacial coupling and the formation of chemical bonds, predominantly Ti-O-C linkages, between the NPC framework and MXene nanosheets.

2.3 Electrode Fabrication

The patch features three 3mm working electrodes for glucose, uric acid, and Na^+ ion sensing, with a shared counter and reference electrode. Flexible electrodes were fabricated from 125μm PI film using CO_2 laser ablation to create 3D porous graphene (LIG).The LIG electrodes were coated with 3D-NPC@2D-$Ti_3C_2T_x$ and heated at 55°C for 10 hours. The glucose electrode was functionalized with Pt, GOx, and Nafion; the uric acid electrode with Pt, Uricase, and Nafion; the Na^+ electrode with Na^+ ISM; and the reference

electrode with Ag and Cl plating.

3. Experimental Results and Discussion

Figure 2(A-F) summarizes the in vitro electrochemical performance of the 3D-NPC@2D-$Ti_3C_2T_x$-modified glucose and uric acid sensors. The glucose and uric acid sensors exhibit high sensitivities of 168 and 226 $\mu AmM^{-1}cm^{-2}$, respectively (Figs. 2A and 2D), along with excellent linearity (Figs. 2B and 2E) and strong selectivity against common interferents (Figs. 2C and 2F). Additionally, the Na^+ ISE shows a stable OCP response up to 160 mM Na^+ (Fig. 3A) with a near-Nernstian sensitivity of 56.96 mV/decade and excellent linearity (R^2 = 0.998; Fig. 3B).

4. Conclusion

This study demonstrates the use of 3D-NPC@2D-$Ti_3C_2T_x$ nanocomposite to modify the LIG electrodes for more efficiently sweat glucose and uric acid sensing. The 3D-NPC@2D-$Ti_3C_2T_x$ /LIG patch sensor enables the simultaneous detection of sweat glucose, uric acid, and Na^+ levels, offering a comprehensive approach to sweat analysis.

Acknowledgments

This research was supported by the Technology Innovation Program (RS-2022-00154983, Development of Low-Power Sensors and Self-Charging Power Sources for Self-Sustainable Wireless Sensor Platforms) funded by the Ministry of Trade, Industry and Energy (MOTIE) and by the Research Program through the Korea Institutes of Police Technology (KIPoT) funded by the Korean National Police Agency & Korea Customs Service (RS-2024-00406234, Development of Field Technology for Responding to Illegal Drugs). The authors are also grateful to the Advanced Sensor and Energy Research (ASER) Laboratory group members of Kwangwoon University.

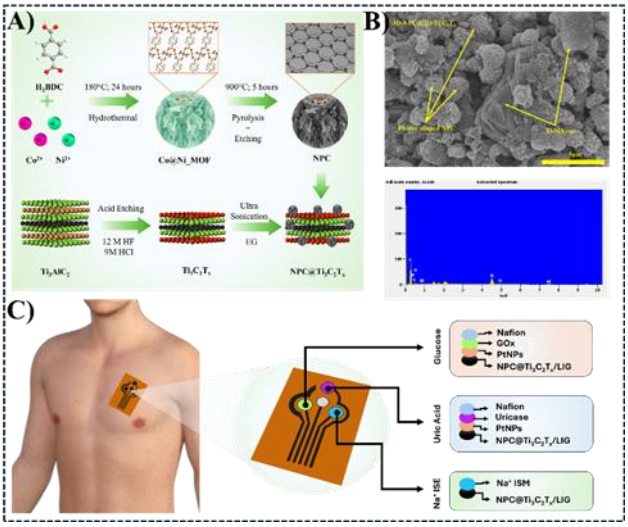

Fig.1. A) Schematic illustration of the synthesis process of 3D-NPC@2D-$Ti_3C_2T_x$. B) FESEM image with EDS spectrum of 3D-NPC@2D-$Ti_3C_2T_x$. C) Schematic illustration of the proposed epidermal patch comprised of three biosensors and different layers of the functionalized electrodes.

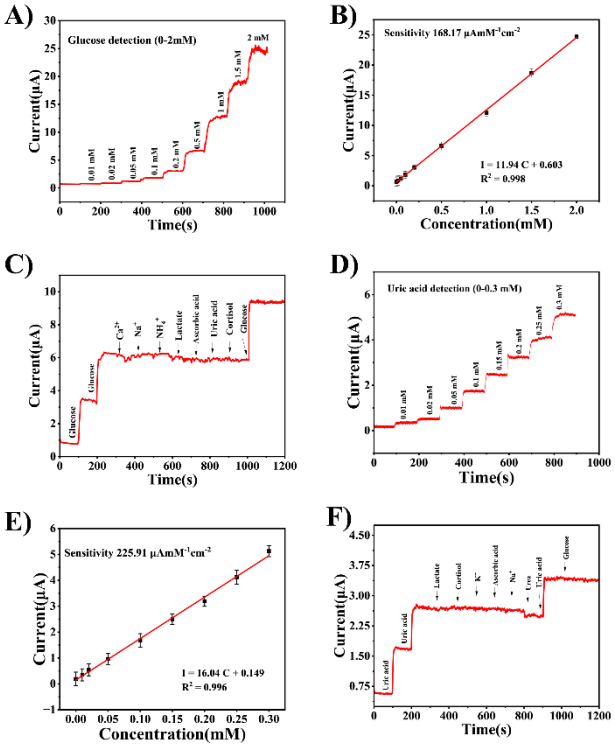

Fig.2. A) Amperometric response of glucose sensor in PBS at 27°C and pH . B) Linearity curve of the glucose sensor (n = 3). C) Selectivity test of the glucose sensor with interfering species. D) Amperometric response of uric acid sensor. E) Linearity curve of the uric acid sensor (n = 3). F) Selectivity test of the uric acid sensor with interfering species.

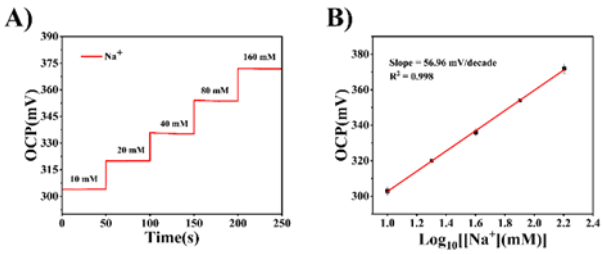

Fig.3. A) OCP response of Na ISE sensor to 160 mM. B) Linear calibration plot of Na ISE (n = 3).

References

[1] J. Kim, A. S. Campbell, B. E. F. de Ávila, J. Wang, *Nat. Biotechnol. 2019 374* **2019**, *37*, 389.

[2] C. Liu, H. Zhao, S. M. Mugo, H. Da Wang, Y. Sun, Q. Zhang, *Adv. Sustain. Syst.* **2026**, *10*, e01068.

[3] M. Asaduzzaman, O. Faruk, A. Abdus Samad, H. Kim, M. Selim Reza, Y. Lee, J. Yeong Park, M. Asaduzzaman, O. Faruk, A. A. Samad, H. Kim, M. S. Reza, Y. Lee, J. Y. Park, *Adv. Funct. Mater.* **2024**, *34*, 2405651.